

ROSHAN RAVI

Greater Boston, MA | [+1 \(617\) 817-5673](tel:+16178175673) | roshan@roshanravi.com | [linkedin.com/in/roshan-ravi](https://www.linkedin.com/in/roshan-ravi)

Work Experience

Innovate Loop

Founder & CEO (September 2019 – Present)

Software Consultant (MongoDB), Major Health Insurance Firm (February 2021 – Present)

- Developed Python/MongoDB ETL platform to handle comprehensive medical data for 4M members
- Optimized database infrastructure to process daily incremental load for 300,000 members
- Leveraged multi-node Python cluster to efficiently parse large XML files into MongoDB within service-level agreement (SLA) of 4 hours
- Implemented transformation mappings for HL7 C-CDA & FHIR data
- Designed ETL platform architecture and presented to key stakeholders in committee review

Nuance Communications

Software Engineer (July 2018 – April 2021)

- Worked on Dragon Medical Virtual Assistant (DMVA), using natural language understanding (NLU) to help streamline clinical workflows via conversational AI
- Architected Azure Kubernetes Service (AKS) cloud deployments
- Converted ASP.NET Core (C#) application into containerized microservices in Docker
- Developed JavaScript/iOS/Android SDKs and sample applications for our REST & WebSocket APIs
- Redesigned internal CLI & IDE tools to simplify the NLU model development process
- Coordinated the integration of our solution within partner applications

Software Engineer Intern (Summer 2017)

- Developed speaker diarization technology to effectively capture doctor-patient conversations
- Built initial prototype of the company's flagship Dragon Ambient Experience (DAX) offering

Software Engineer Intern (Summer 2016)

- Created an application for Nuance's diagnostic division to assist doctors in writing PowerScribe Clinical Guidance Rules to aid creating radiology reports

The Green Alliance (FIRST Robotics Competition Match Analysis)

Founder & Lead Developer (September 2016 – June 2019)

- Led development team to implement a match analysis platform built on MongoDB within 6 weeks
- Coordinated dozens of teams internationally to collect data about 500+ robots over 6000+ matches

Education

University of Massachusetts Lowell – B.S. Computer Science, GPA: 3.975

Fall 2019 – Fall 2021: Data Science Concentration, Math Minor, Commonwealth Honors Scholar, Dean's List

Relevant Coursework: Computing for Health and Medicine (Graduate), Operating Systems (Graduate), Machine Learning, Reinforcement Learning, Analysis of Algorithms

Activities and Societies: Vice President – UML Entrepreneurial Ventures Association (EVA), RiverHawk Robotics, Institute of Electrical and Electronics Engineers (IEEE), Association for Computing Machinery (ACM)

Harvard University Extension School

Summer 2017: Advanced Mobile Application Development Using Swift and iOS

Certifications & Credentials

MongoDB Certified Developer (C100DBA)	See Credential	2021
MongoDB Certified Developer (C100DEV)	See Credential	2021
Certified Kubernetes Administrator (CKA)	See Credential	2020
Certified Kubernetes Application Developer (CKAD)	See Credential	2019
Udacity Full Stack Developer Nanodegree	See Credential	2016
Udacity Front-End Web Developer Nanodegree	See Credential	2016
Udacity iOS Developer Nanodegree	See Credential	2016

Research Experience

University of Massachusetts Lowell

Predicting Outpatient Stroke Rehabilitation Dose-Response (2019 – Present)

- Created classification approach to determine effectiveness of physical therapy for participants recovering from a stroke
- Developed algorithm to predict if patient receiving from stroke will respond positively to our treatment process (accurately predicted for 84% of participants in clinical trials)
- Advised by Prof. Wenjin Zhou (UML) and Prof. Lynne Gauthier (UML)
- Research paper in progress

Synthesizing Crowdsourced Physician Reviews (2020 – 2022)

- Created dashboard to consolidate physician ratings to aid patients in choosing a doctor
- Used PCA & sentimental analysis to provide additional insights
- Advised by Prof. Luvai Motiwalla (UML) and Prof. Nima Kordzadeh (WPI)

Brahmam Innovations Lab (2017 – Present)

- Developing technology solutions to address global challenges
- Advised by Sampathkumar Veeraraghavan (IEEE)

Technical Skills

Orchestration	Kubernetes, Helm, Docker, Docker Compose, Nginx
Cloud Providers	Microsoft Azure, Amazon Web Services, Google Cloud Platform, DigitalOcean
Languages	Node.JS, C#, Python, C++, Go, TypeScript, JavaScript, Swift, Java, C, MATLAB
Data Analysis	TensorFlow/Keras, PyTorch, Jupyter, D3.js, Tableau
Databases	MongoDB, Redis, MySQL, Google Firebase
CI/CD	Git, GitHub, Azure DevOps, Atlassian Bitbucket

Awards & Honors

Manning BEST (Business & Entrepreneur Scholars in Training)	2021
Kennedy Colleges of Sciences Research Scholar	2020
David E. Shulman Computer Science ACM Leadership Award	2020
UMass Lowell Immersive Scholar, Dean's Scholarship	2019
Presenter – FIRST Robotics World Championship Conference in Detroit	2018
FIRST Robotics Dean's List Award Semi-Finalist	2018